

Unit 7

Coordinate Geometry

Sr. No.	Questions	A	B	C	D
1	The equation of a straight line in the slope-intercept form is written as:	$y = m(x + c)$	$y - y_1 = m(x - x_1)$	$y = c + mx$ ✓	$ax + by + c = 0$
2	The gradients of two parallel lines are:	equal✓	zero	negative reciprocals of each other	always undefined
3	If the product of the gradients of two lines is -1 , then the lines are:	parallel	perpendicular✓	collinear	coincident
4	Distance between two points $P(1,2)$ and $Q(4,6)$ is:	5✓	6	$\sqrt{13}$	4
5	The midpoint of a line segment with endpoints $(-2,4)$ and $(6,-2)$ is:	$(4,2)$	$(2,1)$ ✓	$(1,1)$	$(0,0)$
6	A line passing through points $(1,2)$ and $(4,5)$ is:	$y = x + 1$ ✓	$y = 2x + 3$	$y = 3x - 2$	$y = x + 2$
7	The equation of a line in point-slope form is:	$y = m(x + c)$	$y - y_1 = m(x - x_1)$ ✓	$y = c + mx$	$ax + by + c = 0$
8	$2x + 3y - 6 = 0$ in the slope-intercept form is:	$y = \frac{-2}{3}x + 2$ ✓	$y = \frac{2}{3}x - 2$	$y = \frac{2}{3}x + 1$	$y = \frac{-2}{3}x - 2$
9	The equation of a line in symmetric form is:	$\frac{x}{a} + \frac{y}{b} = 1$	$\frac{x - x_1}{1} + \frac{y - y_1}{m} = \frac{z - z_1}{1}$	$\frac{x - x_1}{\cos \alpha} = \frac{y - y_1}{\sin \alpha} = r$ ✓	$y - y_1 = m(x - x_1)$
10	The equation of a line in normal form is:	$y = mx + c$	$\frac{x}{a} + \frac{y}{b} = 1$	$\frac{x - x_1}{\cos \alpha} = \frac{y - y_1}{\sin \alpha}$	$x \cos \alpha + y \sin \alpha = p$ ✓

Solution of MCQs

1	Slope-Intercept Form is $y = c + mx$ Also written as $y = mx + c$.
2	Parallel lines have equal slopes .
3	Perpendicular lines: product of slopes = -1
4	Distance between $P(1,2)$ and $Q(4,6)$ $ PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $= \sqrt{(4 - 1)^2 + (6 - 2)^2}$ $= \sqrt{(3)^2 + (4)^2}$ $= \sqrt{9 + 16}$ $= \sqrt{25}$ $= 5$

5	Midpoint of $(-2, 4)$ and $(6, -2)$: $M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$ $= \left(\frac{-2 + 6}{2}, \frac{4 + (-2)}{2} \right)$ $= \left(\frac{4}{2}, \frac{2}{2} \right)$ $= (2, 1)$
6	Slope of line from $(1, 2)$ to $(4, 5)$: $m = \frac{y_2 - y_1}{x_2 - x_1}$ $m = \frac{5 - 2}{4 - 1}$ $m = \frac{3}{3}$ $m = 1$ Now Point Slope Form $y - y_1 = m(x - x_1)$ $y - 2 = 1(x - 1)$ $y = x - 1 + 2$ $y = x + 1$
7	Equation of a line in point-slope form is $y - y_1 = m(x - x_1)$
8	$2x + 3y - 6 = 0$ $\Rightarrow 3y = -2x + 6$ $y = \frac{-2x + 6}{3}$ $y = \frac{-2x}{3} + \frac{6}{3}$ $y = \frac{-2x}{3} + 2$
9	$\frac{x - x_1}{\cos \alpha} = \frac{y - y_1}{\sin \alpha} = r$
10	$x \cos \alpha + y \sin \alpha = P$